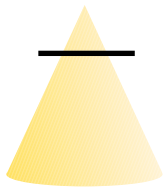
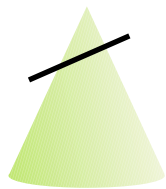


CUTTING CONES

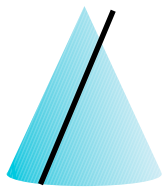
An ancient Greek mathematician called Appolonius discovered that you can create the most important mathematical curves simply by cutting through a cone shape. The four most important types of curve are shown below.



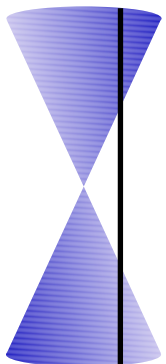
If you make a horizontal cut straight through a cone, the cut forms a **circle**.



Cutting across the cone at an angle produces a kind of oval called an **ellipse**.

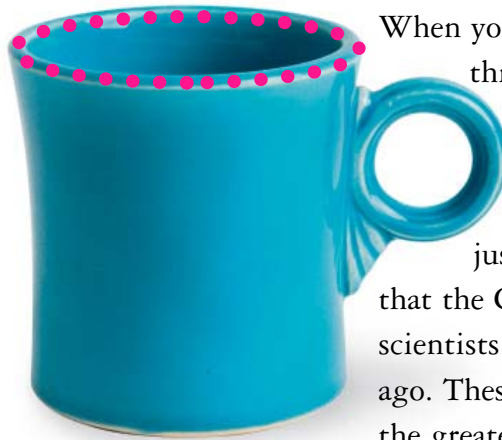


If you cut parallel to the side of the cone, the curve is a kind of arch called a **parabola**.



If you cut straight down through two cones placed tip to tip, you get twin curves – a pattern called a **hyperbola**.

Cones *and* curves



Whenever you look at a circular object, you nearly always see it as an *ellipse*.

When you throw a ball, whether you throw it high or long, it will always fly in a curve called a *parabola* and arrive back on Earth. The parabola is just one of the maths curves that the Greeks discovered and that scientists began to explore 400 years ago. These explorations led to one of the greatest discoveries of all time: that everything in the Universe pulls on everything else through the mysterious force of *gravity*.

WHAT'S A PARABOLA?

When a football, a leaping dolphin, a waterfall, or a cannon ball flies through the air, it travels along a curved path. The great Italian scientist Galileo discovered this curve was a parabola. He realized that a cannon ball flies horizontally at a *constant speed* (allowing for air resistance), but the continuous pull of gravity makes it fall vertically at an *accelerating speed*. The result is a curve that gets steeper and steeper – a parabola.

Why doesn't the

Galileo changed the world, for suddenly people stopped building city walls. For the first time, gunners could fire over walls and knew exactly where a cannon ball would land. A man called Kepler then found that planets revolve around the Sun in ellipses, and the genius scientist Isaac Newton saw that the Moon behaved in the same way as a cannon ball. So why doesn't the Moon fall to Earth?